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Inventor(s)	Yoneyama; Yojiro

Wind direction adjustment apparatus

Abstract

Provided is a wind direction adjustment apparatus achieving improved performance while the number of components is reduced. A wind direction adjustment apparatus **1** includes a case body **3** that internally defines a ventilation passage **5**. The wind direction adjustment apparatus **1** includes a fin **15** pivotably disposed in the ventilation passage **5** in the case body **3** to adjust a wind direction in accordance with pivoting. The wind direction adjustment apparatus **1** includes a valve **30** pivotably disposed in a direction intersecting with a pivoting direction of the fin **15** in the ventilation passage **5** in the case body **3**, and capable of opening and closing the ventilation passage **5** in accordance with the pivoting. The wind direction adjustment apparatus **1** includes a link **35** that interlocks the pivoting of the valve **30** with the pivoting of the fin **15**.

Inventors:	Yoneyama; Yojiro (Shizuoka, JP)
Applicant:	NIHON PLAST CO., LTD. (Fujinomiya, JP)
Family ID:	1000008777533
Assignee:	NIHON PLAST CO., LTD. (Fujinomiya, JP)
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Primary Examiner: Savani; Avinash A

Assistant Examiner: Tighe; Dana K

Attorney, Agent or Firm: Amin, Turocy & Watson, LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) The disclosure of Japanese Patent Application No. 2021-194126 filed on Nov. 30, 2021 including the specification, drawings and abstract is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The present invention relates to a wind direction adjustment apparatus including a valve capable of opening and closing a ventilation passage.

BACKGROUND ART

(3) Conventionally, in an air conditioner used in a vehicle such as an automobile, a wind direction adjustment apparatus that adjusts a blowing wind direction is known. The wind direction adjustment apparatus is also called an air-conditioned wind blowing apparatus, an air outlet, a

ventilator, or a register. For example, the wind direction adjustment apparatus is installed in various components of the vehicle, such as an instrument panel and a center console part, and contributes to improvement in comfort performance achieved by cooling and heating.

(4) In the wind direction adjustment apparatus configured in this way, there is known an apparatus that includes a fin for adjusting a wind direction and a shut valve for opening and closing a ventilation passage. In the apparatus, the fin is operated to pivot by an operation knob, and the shut valve is operated by an operation dial (for example, refer to PTL 1).

(5) In a case of this configuration, the operation knob and the operation dial are required. Consequently, a mechanism is complicated, layout restrictions are imposed, and design is likely to be impaired.

(6) In addition, in place of the shut valve, a technique is known in which a plurality of fins overlap each other at a maximum swung position to close the ventilation passage (for example, refer to PTL 2).

(7) In a case of this configuration, it is not easy to hold a sealed state of the ventilation passage only by overlapping the fins. This configuration requires a structure in which a surface of the fin is provided with adhesion, or a structure including a mechanism that biases the fin in a pressing direction.

CITATION LIST

Patent Literature

(8) PTL 1 Japanese Patent Application Laid-Open No. 2007-106352 (pages 4 to 6, FIG. 1) PTL 2 Japanese Patent Application Laid-Open No. 2010-89529 (pages 9 to 10, FIG. 7)

SUMMARY OF INVENTION

Technical Problem

(9) As described above, it is desired to improve performance of the wind direction adjustment apparatus with a simpler configuration.

(10) The present invention is made in view of the above-described circumstances, and an object of the present invention is to provide a wind direction adjustment apparatus achieving improved performance while the number of components is reduced.

Solution to Problem

(11) There is provided a wind direction adjustment apparatus according to claim 1. The wind direction adjustment apparatus includes a case body that internally defines a ventilation passage, a fin disposed in the ventilation passage in the case body to be pivotable to adjust a wind direction in accordance with pivoting, a valve disposed in the ventilation passage in the case body to be pivotable in a direction intersecting with a pivoting direction of the fin, and capable of opening and closing the ventilation passage in accordance with pivoting, and a link that interlocks the pivoting of the valve with the pivoting of the fin.

(12) According to the wind direction adjustment apparatus of claim 2, the wind direction adjustment apparatus of claim 1 further includes an operation section that causes the fin to pivot. The fin may be pivotable in an operation direction of the operation section.

(13) According to the wind direction adjustment apparatus of claim 3 or 4, in the wind direction adjustment apparatus of claim 1 or 2, the link may be movable in a direction along the pivoting direction of the fin. The valve may have a link joining portion joined to the link, and may be caused to pivot by an external force applied to the link joining portion from the link side as the link moves.

(14) According to the wind direction adjustment apparatus of the claims, a plurality of the fins may be disposed in a longitudinal direction of the case body. The valve may be disposed in a longitudinal shape along the longitudinal direction.

Advantageous Effects of Invention

(15) According to the wind direction adjustment apparatus of claim 1, for example, it is not necessary to separately provide the operation section for operating the fins and the operation section for operating the valve. While the number of components can be reduced, sealing of the

ventilation passage can be easily held by the valve. Therefore, performance can be improved.

(16) According to the wind direction adjustment apparatus of claim 2, in addition to an advantageous effect of the wind direction adjustment apparatus of claim 1, the fin can be intuitively operated by an operation of the operation section, and the fin can pivot in accordance with the operation of the operation section. The valve can pivot by being interlocked with the fin. The fin and the valve can be operated by using a common operation section. Therefore, a configuration can be simplified.

(17) According to the wind direction adjustment apparatus of claim 3 or 4, in addition to an advantageous effect of the wind direction adjustment apparatus of claim 1 or 2, the pivoting of the valve can be easily interlocked with the pivoting of the fin by the link.

(18) According to the wind direction adjustment apparatus according to the claims, the wind direction adjustment apparatus having a thin shape can be configured.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a perspective view showing an internal structure of a wind direction adjustment apparatus according to embodiment 1 of the present invention;
- (2) FIG. 2 is an exploded perspective view showing the internal structure of the wind direction adjustment apparatus;
- (3) FIGS. 3A to 3D are plan views showing operations of a fin and a valve of the wind direction adjustment apparatus, FIG. 3A shows a state where the fin of the wind direction adjustment apparatus is swung to the maximum in one direction, FIG. 3B shows a neutral state of the fin, FIG. 3C shows a state where the fin is swung in the other direction, and FIG. 3D shows a closed state of the valve;
- (4) FIG. 4 is a perspective view showing the wind direction adjustment apparatus;
- (5) FIG. 5 is a perspective view showing an internal structure of a wind direction adjustment apparatus according to embodiment 2 of the present invention;
- (6) FIG. 6 is an exploded perspective view showing the internal structure of the wind direction adjustment apparatus;
- (7) FIG. 7 is a perspective view showing an enlarged part of the internal structure;
- (8) FIGS. 8A to 8D are plan views showing operations of a fin and a valve of the wind direction adjustment apparatus, FIG. 8A shows a state where the fin of the wind direction adjustment apparatus is swung to the maximum in one direction, FIG. 8B shows a neutral state of the fin, FIG. 8C shows a state where the fin is swung in the other direction, and FIG. 8D shows a closed state of the valve; and
- (9) FIG. 9 is a perspective view showing the wind direction adjustment apparatus.

DESCRIPTION OF EMBODIMENTS

- (10) Hereinafter, embodiment 1 of the present invention will be described with reference to the drawings.
- (11) In FIG. 4, a reference number 1 represents a wind direction adjustment apparatus. The wind direction adjustment apparatus 1 is also called an air outlet, a ventilator, or a register, and adjusts a wind blowing direction from an air conditioner. Hereinafter, for more clear description, in the wind direction adjustment apparatus 1, a leeward side from which wind blows will be defined as a front side, a front surface side, or a near side. A side opposite thereto, that is, a windward side which receives the wind will be defined as a rear side, a back side, or a far side. In this manner, a both side direction or a width direction which is a rightward-leftward direction when viewed from the front side, and an upward-downward direction will be defined. In the present embodiment, the wind direction adjustment apparatus 1 is applied to an air conditioner for a vehicle such as an

automobile. The wind direction adjustment apparatus **1** may be disposed at any desired position. In the drawings, an arrow FR side will be defined as the front side, an arrow RR side will be defined the rear side, an arrow L side will be defined as a left side, an arrow R side will be defined as a right side, an arrow U side will be defined as an upper side, and an arrow D side will be defined as a lower side. The directions are shown only as examples, and may be changed as appropriate depending on an installation position or an installation orientation of the wind direction adjustment apparatus **1**.

(12) The wind direction adjustment apparatus **1** includes a case body **3**. The case body **3** is also called a duct. The case body **3** is formed in a cylindrical shape. In the present embodiment, the case body **3** is formed in a cylindrical shape in a forward-rearward direction. In the shown example, the case body **3** is formed in a rectangular cylindrical shape. A ventilation passage **5** is internally surrounded by the case body **3**. A direction parallel to a central axis of the case body **3** is a ventilation direction of the ventilation passage **5**. In the present embodiment, the ventilation direction of the ventilation passage **5** is the forward-rearward direction, and ventilation is performed from the rear side to the front side. That is, in the ventilation passage **5**, the rear side is an upstream side in the ventilation direction, and the front side is a downstream side in the ventilation direction.

(13) The case body **3** has a predetermined length in the ventilation direction of the ventilation passage **5**. In the present embodiment, the case body **3** is flat in an upward-downward direction, and is formed in a longitudinal shape in the rightward-leftward direction, that is, in a horizontally long shape. Therefore, the wind direction adjustment apparatus **1** is formed in a horizontally thin shape. The case body **3** integrally has a central portion of the ventilation passage **5**, that is, a pair of end wall portions **6** facing each other across the central axis, and a pair of side wall portions **7** joining the pair of end wall portions **6**. The pair of end wall portions **6** face each other in the upward-downward direction, and the pair of side wall portions **7** face each other in the rightward-leftward direction. A receiving port **8** for receiving air, that is, air-conditioned wind in the ventilation passage **5**, is surrounded by rear end portions of the pair of end wall portions **6** and **6** and the pair of side wall portions **7** and **7**. An air outlet **9** for discharging the air-conditioned wind from the ventilation passage **5** is surrounded by front end portions of the pair of end wall portions **6** and **6** and the pair of side wall portions **7** and **7**. That is, the rear end portion of the case body **3** serves as the receiving port **8** for receiving the air-conditioned wind into the ventilation passage **5**, and the front end portion of the case body **3** serves as the air outlet **9** for discharging the air-conditioned wind from the ventilation passage **5**. The ventilation passage **5** allowing communication is formed between the receiving port **8** and the air outlet **9**. The air-conditioned wind passes from the receiving port **8** to the air outlet **9**. The receiving port **8** and the air outlet **9** respectively have the horizontally long shape.

(14) The case body **3** may be integrally formed, or may be formed by combining a plurality of members. In the present embodiment, the case body **3** has a case main body portion **11** and a finisher **12** which is a design member. The case main body portion **11** is a main body portion forming an upstream side majority of the case body **3**. The case main body portion **11** is formed in a rectangular cylindrical shape. The finisher **12** is attached to a front end, that is, a downstream end of the case main body portion **11**. The finisher **12** is also called a panel, and forms a portion of the design at an installation position of the wind direction adjustment apparatus **1**. The finisher **12** is formed in a rectangular frame shape surrounding the air outlet **9**.

(15) As shown in FIG. **1**, a fin **15** is disposed inside the case body **3**, that is, in the ventilation passage **5**. The fin **15** is also called a louver, and pivots with respect to the case body **3**, thereby adjusting a wind direction of the air-conditioned wind blowing from the air outlet **9** (FIG. **4**) in accordance with the pivoting. The fin **15** is formed in a plate shape in which one main surface and the other main surface serve as flow straightening surfaces. The fin **15** has a pivoting portion **16**. The pivoting portion **16** is held to be pivotable by a pivoting receiving portion **17** (FIG. **4**) formed

in the case body **3**. The fin **15** pivots along a longitudinal direction of the case body **3** or the ventilation passage **5** to adjust the wind direction in the longitudinal direction of the case body **3** or the ventilation passage **5**. That is, in the present embodiment, the fin **15** has the pivoting portions **16** on upper and lower sides, and each pivoting portion **16** is held to be pivotable by the pivoting receiving portion **17** (FIG. **4**) formed on each end wall portion **6** of the case body **3**. The fin **15** has flow straightening surfaces on right and left sides, and is pivotable in the rightward-leftward direction. One of the pivoting portion **16** and the pivoting receiving portion **17** (FIG. **4**) is a shaft portion, and the other is a hole portion or a recessed portion. In the present embodiment, the pivoting portion **16** is the shaft portion, and the pivoting receiving portion **17** (FIG. **4**) is the hole portion having a round hole shape or the recessed portion.

(16) In the present embodiment, the fin **15** is located inside the case main body portion **11** in the case body **3**. That is, the fin **15** is located to be separated to an upstream side with respect to the air outlet **9**. The pivoting receiving portion **17** is formed in the case main body portion **11**.

(17) The number of fins **15** may be one or two or more. In the present embodiment, a plurality of the fins **15** are disposed in the longitudinal direction of the case body **3**. Preferably, the plurality of fins **15** are disposed at an equal interval or an approximately equal interval in the longitudinal direction of the case body **3**. In the present embodiment, the fin **15** are aligned in the rightward-leftward direction. The plurality of fins **15** are joined by a link member, and configured to pivot in the same direction by being interlocked with each other. In the drawing, for clear description, only one fin **15** disposed in a central portion is shown, and illustration of the other fins **15** is omitted.

(18) As shown in FIGS. **1** and **2**, in the present embodiment, the fin **15** is joined to the operation section **20**, and the pivoting of the fin **15** can be directly operated by a user such as an occupant through the operation section **20**. When there are the plurality of fins **15**, one of the fin **15**, preferably the fin **15** in the central portion is joined to the operation section **20**. The operation section **20** is an operation knob, and the fin **15** is pivotable in an operation direction of the operation section **20**. In the shown example, the operation section **20** is movable in the rightward-leftward direction, and the fin **15** is caused to pivot in the rightward-leftward direction by the movement in the rightward-leftward direction. The operation section **20** is exposed from the air outlet **9** (FIG. **4**). In the present embodiment, the operation section **20** is formed in a longitudinally thin shape in the rightward-leftward direction. A joining portion **21** is formed in the operation section **20**. The joining portion **21** is joined to a joining receiving portion **22** formed in the fin **15** to be pivotable. One of the joining portion **21** and the joining receiving portion **22** is a shaft portion, and the other is a hole portion or a recessed portion. In the present embodiment, the joining portion **21** is a recessed portion which is long in the forward-rearward direction between a pair of arm portions projecting from a rear portion of the operation section **20** toward the fin **15**, and the joining receiving portion **22** is a columnar shaft portion formed parallel or substantially parallel to a pivoting axis in the fin **15**.

(19) The operation section **20** is movably attached to a downstream side fin **24** which is an operation guide portion. In the shown example, the operation section **20** is attached to be slidable along the downstream side fin **24**. The downstream side fin **24** is also called a downstream side louver. The downstream side fin **24** is formed in a plate shape in which one main surface and the other main surface serve as flow straightening surfaces. The downstream side fin **24** is disposed to have the straightening surface in a direction intersecting with or orthogonal to the fin **15**, on the downstream side from the fin **15**. In the present embodiment, the downstream side fin **24** is disposed to have the flow straightening surfaces on upper and lower sides. The downstream side fin **24** is disposed in a longitudinal shape along the longitudinal direction of the case body **3**. A restriction portion **25** that restricts a movement range of the operation section **20** is formed in the downstream side fin **24**. The restriction portion **25** is a stopper portion that does not allow the operation section **20** to move any further by coming into contact with the operation section **20**. The restriction portion **25** is formed on a back side of the downstream side fin **24**. In the present

embodiment, a pivotable range of the fin **15** by the operation section **20** is set by the restriction portion **25** to be wider in the rightward direction than in the leftward direction.

(20) Preferably, the downstream side fin **24** is disposed to be pivotable in the case body **3** in a direction intersecting with or orthogonal to the pivoting direction of the fin **15**. The downstream side fin **24** has a pivoting portion **26**. The pivoting portion **26** is held to be pivotable by a pivoting receiving portion formed in the case body **3**. The downstream side fin **24** pivots along a direction intersecting with or orthogonal to the longitudinal direction of the case body **3** or the ventilation passage **5**, thereby adjusting the wind direction in the direction intersecting with or orthogonal to the longitudinal direction of the case body **3** or the ventilation passage **5**. In the shown example, the downstream side fin **24** is disposed to be pivotable in the upward-downward direction. That is, in the present embodiment, the downstream side fin **24** has pivoting portions **26** on right and left sides, and each of the pivoting portions **26** is held to be pivotable by a pivoting receiving portion formed in each side wall portion **7** of the case body **3**. The downstream side fin **24** has the flow straightening surfaces on upper and lower sides, and is pivotable in the upward-downward direction. One of the pivoting portion **26** and the pivoting receiving portion is a shaft portion, and the other is a hole portion or a recessed portion. In the present embodiment, the pivoting portion **26** is the shaft portion, and the pivoting receiving portion is the hole portion having a round hole shape or the recessed portion. The downstream side fin **24** is pivotable along the movement direction of the operation section **20**. That is, the downstream side fin **24** pivots up and down integrally with the operation section **20** as the operation section **20** moves up and down.

(21) In the present embodiment, as shown in FIG. **4**, the downstream side fin **24** is located inside the finisher **12** in the case body **3**. That is, the downstream side fin **24** is located to face the air outlet **9**. The pivoting receiving portion is formed in the finisher **12**. In the shown example, one downstream side fin **24** is set in the central portion in the upward-downward direction which is a short direction of the air outlet **9**.

(22) In addition, a valve **30** shown in FIGS. **1** and **2** is disposed inside the case body **3**, that is, in the ventilation passage **5**. The valve **30** is a shut valve that pivots with respect to the case body **3** to open and close the ventilation passage **5** in accordance with the pivoting. The valve **30** is formed in a rectangular plate shape. The valve **30** has an outer shape substantially equal to a cross-sectional shape of the ventilation passage **5**. The valve **30** has a valve pivoting portion **31**. The valve pivoting portion **31** is held to be pivotable by a valve pivoting receiving portion **32** (FIG. **4**) formed in the case body **3**. The valve **30** opens and closes the ventilation passage **5** by pivoting in the direction intersecting with the longitudinal direction of the case body **3** or the ventilation passage **5**. That is, in the present embodiment, the valve **30** has valve pivoting portions **31** on right and left sides, and each of the valve pivoting portions **31** is held to be pivotable by the valve pivoting receiving portion **32** (FIG. **4**) formed in each side wall portion **7** of the case body **3**. The valve **30** is pivotable in the upward-downward direction. Therefore, the pivoting direction of the valve **30** intersects with or is orthogonal to the pivoting direction of the fin **15**. One of the valve pivoting portion **31** and the valve pivoting receiving portion **32** (FIG. **4**) is a shaft portion and the other is a hole portion or a recessed portion. In the present embodiment, the valve pivoting portion **31** is the shaft portion, and the valve pivoting receiving portion **32** (FIG. **4**) is the hole portion having a round hole shape or the recessed portion.

(23) The valve **30** is located to be separated to the upstream side with respect to the fin **15**. In the present embodiment, the valve **30** is located inside the case main body portion **11** and in the vicinity of the receiving port **8** (FIG. **4**) in the case body **3**. The valve pivoting receiving portion **32** (FIG. **4**) is formed in the case main body portion **11**.

(24) The pivoting of the valve **30** is interlocked with the pivoting of the fin **15** via a link **35**. The link **35** is directly or indirectly joined to the valve **30** and the fin **15** to be pivotable. In the present embodiment, the link **35** is directly joined to each of the valve **30** and the fin **15** to be pivotable. The link **35** is movable in a direction along the pivoting direction of the fin **15**, and is configured to

cause the valve **30** to pivot by transmitting an external force to the valve **30** side as the link **35** moves.

(25) The link **35** is formed in a longitudinal shape. The link **35** is disposed in the ventilation passage **5** to have the longitudinal direction in the forward-rearward direction. The link **35** has a rear end portion in which a valve side joining portion **37** joined to the valve **30** side is one end portion. In the present embodiment, the valve side joining portion **37** is directly joined to a link joining portion **38** formed in the valve **30**. The valve side joining portion **37** is joined to the link joining portion **38** to be pivotable and movable. The valve side joining portion **37** and the link joining portion **38** form an interlocking portion that interlocks opening and closing of the valve **30** with the pivoting of the fin **15**, that is, the operation of the operation section **20**. One of the valve side joining portion **37** and the link joining portion **38** is a shaft portion, and the other is a hole portion. The valve side joining portion **37** is formed along the pivoting direction of the fin **15** in a cylindrical shape having a hole portion **37a** penetrating in a tangential direction of the pivoting direction of the fin **15** in the present embodiment. The hole portion **37a** is formed in a long hole shape in the longitudinal direction of the link **35**. A pivoting range (pivoting angle) of the fin **15** in which an external force for causing the valve **30** to pivot is generated by the pivoting of the fin **15**, that is, an operation range of the operation section **20** is set according to the length of the hole portion **37a**. That is, an opening/closing timing of the valve **30** is set according to the length of the hole portion **37a**. The hole portion **37a** is a timing setting portion that sets a timing for interlocking the pivoting of the fin **15** and the pivoting of the valve **30** with each other. Since the hole portion **37a** is set, the valve **30** is configured to pivot by being interlocked with the fin **15** when the fin **15** is swung at a predetermined angle or larger. In addition, in the present embodiment, the valve side joining portion **37** has a cutout opening portion **37b** through which the hole portion **37a** and the outside of the valve side joining portion **37** communicate with each other. The cutout opening portion **37b** can join the valve side joining portion **37** and the link joining portion **38** to each other. The cutout opening portion **37b** has a slit shape extending in the rightward-leftward direction.

(26) The link joining portion **38** is formed in a longitudinal shape along the longitudinal direction of the valve **30**. The link joining portion **38** is formed in an elongated columnar shape or a rod shape. The valve side joining portion **37** of the link **35** is movable along the longitudinal direction of the link joining portion **38**, and the link joining portion **38** is pivotable with respect to the valve side joining portion **37**. The link joining portion **38** is disposed parallel or substantially parallel to the pivoting axis on a side opposite to the link **35** with reference to the pivoting axis of the valve **30**, that is, on the upstream side with respect to the pivoting axis of the valve **30**. Both end portions of the link joining portion **38** are fixed to one main surface of the valve **30** by a fixing portion **39**. Therefore, the valve **30** pivots in such a manner that a force is applied from the link **35** to the link joining portion **38** in the forward-rearward direction which is an upstream/downstream direction.

(27) In the present embodiment, the link joining portion **38** has a contraction portion **38a** in one end portion in the longitudinal direction to be joined to the valve side joining portion **37**. The contraction portion **38a** is formed to be narrower than other general portions of the link joining portion **38**, and can be inserted or press-fitted into the hole portion **37a** from the cutout opening portion **37b** of the valve side joining portion **37**. Preferably, the contraction portion **38a** is set outside a movement range of the valve side joining portion **37** of the link **35** with respect to the link joining portion **38**.

(28) In addition, the link **35** has a front end portion in which a fin side joining portion **41** joined to the fin **15** side is the other end portion. In the present embodiment, the fin side joining portion **41** is directly joined to a link side joining receiving portion **42** formed in the fin **15**. The fin side joining portion **41** is joined to the link side joining receiving portion **42** to be pivotable. One of the fin side joining portion **41** and the link side joining receiving portion **42** is a shaft portion, and the other is a hole portion. In the present embodiment, the fin side joining portion **41** is formed in a cylindrical shape having a hole portion **41a** penetrating in a direction parallel or substantially parallel to the

pivoting axis of the fin **15**, that is, in the direction intersecting with or orthogonal to the pivoting direction of the fin **15**. The hole portion **41a** has a round hole shape. The link side joining receiving portion **42** is formed in an elongated columnar shape or a rod shape. The link side joining receiving portion **42** is disposed at a position on a side opposite to the joining receiving portion **22** with reference to the pivoting portion **16** of the fin **15**. The link side joining receiving portion **42** is formed parallel or substantially parallel to the pivoting axis of the fin **15**.

(29) The wind direction adjustment apparatus **1** is disposed by joining the receiving port **8** to an air conditioner. The air-conditioned wind from the air conditioner passes through the ventilation passage **5** from the receiving port **8**, is distributed by the fin **15** and the downstream side fin **24**, and blows from the air outlet **9**.

(30) The wind direction adjustment apparatus **1** of the present embodiment can blow the air-conditioned wind in any desired direction by combining the wind distributed by the fin **15** in the rightward-leftward direction and the wind distributed by the downstream side fin **24** in the upward-downward direction.

(31) First, with regard to the wind distribution in the upward-downward direction, when a user such as an occupant grips the operation section **20** and moves the operation section **20** in the upward-downward direction, the downstream side fin **24** pivots up and down integrally with the operation section **20**. In this manner, the air-conditioned wind is straightened in the upward-downward direction along the flow straightening surface of the downstream side fin **24**, and blows in the upward-downward direction from the air outlet **9**.

(32) With regard to the wind distribution in the rightward-leftward direction, a user such as an occupant grips the operation section **20** and operates the operation section **20** to slide rightward and leftward. In this manner, the fin **15** pivots rightward and leftward. The air-conditioned wind is straightened in the rightward-leftward direction along the flow straightening surface of the fin **15**, and blows in the rightward-leftward direction from the air outlet **9**.

(33) Specifically, as shown in FIG. **3B**, when the operation section **20** is located in the central portion of the downstream side fin **24** in the rightward-leftward direction, that is, when the fin **15** is located at a neutral position, the air-conditioned wind straightly flows to the air outlet **9** along the flow straightening surface of the fin **15** inside the ventilation passage **5**, and blows in the axial direction of the case body **3**, that is, in a front surface direction the ventilation direction of the ventilation passage **5** from the air outlet **9**.

(34) In addition, as shown in FIG. **3A**, when the operation section **20** is slid leftward along the downstream side fin **24** from the neutral position, the downstream side of the fin **15** joined to the operation section **20** moves leftward, and the fin **15** pivots in a clockwise direction around the pivoting portion **16** in the drawing. FIG. **3A** shows a state where the operation section **20** is slid leftward to the maximum by being moved until the operation section **20** comes into contact with the restriction portion **25** on the left side, that is, a state where the fin **15** is swung leftward to the maximum (maximum leftward swing angle position, for example, 35° to the left). Therefore, the air-conditioned wind is straightened in the leftward direction along the flow straightening surface of the fin **15** inside the ventilation passage **5**, and blows in the leftward direction from the air outlet **9**.

(35) As the fin **15** pivots in the clockwise direction, the link side joining receiving portion **42** also moves in the clockwise direction around the pivoting portion **16**. Accordingly, in the link **35** in which the fin side joining portion **41** is joined to the link side joining receiving portion **42**, the fin side joining portion **41** pivots in the clockwise direction around the pivoting portion **16**, and the valve side joining portion **37** is joined to the link joining portion **38** of the valve **30**. In this manner, while maintaining a posture along the forward-rearward direction, the link **35** slides in the rightward direction along the link joining portion **38**. In this case, the hole portion **37a** of the valve side joining portion **37** has a long hole shape in the forward-rearward direction. Accordingly, over the entire movement range of the link **35**, the link **35** does not interfere with the link joining portion

38 of the valve **30** inserted into the hole portion **37a** of the valve side joining portion **37** in the forward-rearward direction. Therefore, the valve **30** is not interlocked with the pivoting of the fin **15**, and the valve **30** maintains the open state of the ventilation passage **5**.

(36) On the other hand, as shown in FIG. 3C, when the operation section **20** is slid rightward along the downstream side fin **24** from the neutral position, the downstream side of the fin **15** joined to the operation section **20** moves rightward, and the fin **15** pivots around the pivoting portion **16** in a counterclockwise direction in the drawing. Therefore, the air-conditioned wind is straightened rightward along the flow straightening surface of the fin **15** inside the ventilation passage **5**, and blows in the rightward direction from the air outlet **9**.

(37) As the fin **15** pivots in the counterclockwise direction, the link side joining receiving portion **42** also moves in the counterclockwise direction around the pivoting portion **16**. Accordingly, in the link **35** in which the fin side joining portion **41** is joined to the link side joining receiving portion **42**, the fin side joining portion **41** pivots in the counterclockwise direction around the pivoting portion **16**, and the valve side joining portion **37** is joined to the link joining portion **38** of the valve **30**. In this manner, while maintaining a posture along the forward-rearward direction, the link **35** slides in the leftward direction along the link joining portion **38**. In this case, the hole portion **37a** of the valve side joining portion **37** has a long hole shape in the forward-rearward direction. Accordingly, over a predetermined range (for example, a range in which the fin **15** pivots to the right by 35°), the link **35** does not interfere with the link joining portion **38** of the valve **30** inserted into the hole portion **37a** of the valve side joining portion **37** in the forward-rearward direction. Therefore, the valve **30** is not interlocked with the pivoting of the fin **15**, and the valve **30** maintains the open state of the ventilation passage **5**.

(38) As shown in FIG. 3D, when the operation section **20** is further slid rightward along the downstream side fin **24** from a state shown in FIG. 3C, the link **35** further pivots as the fin **15** further pivots. In this manner, the valve side joining portion **37** interferes with the link joining portion **38** of the valve **30** in the forward-rearward direction, and pulls the link joining portion **38** to the downstream side, that is, to the front side. Therefore, the valve **30** pivots by being interlocked with the pivoting of the fin **15**, and closes the ventilation passage **5** in accordance with a pivoting angle thereof. FIG. 3D shows a state where the operation section **20** is slid rightward to the maximum by being moved until the operation section **20** comes into contact with the restriction portion **25** on the right side, that is, a state where the fin **15** is swung rightward to the maximum (maximum rightward swing angle position, for example, 82° to the right). At least in this state, the valve **30** is located at a position where the ventilation passage **5** is completely or substantially completely closed.

(39) In this way, according to the embodiment 1, the pivoting of the valve **30** disposed in the ventilation passage **5** in the case body **3** to be pivotable in the direction intersecting with the pivoting direction of the fin **15** is interlocked with the pivoting of the fin **15** disposed in the ventilation passage **5** in the case body **3** to be pivotable. Accordingly, for example, it is not necessary to separately provide the operation section for operating the fin **15** and the operation section for operating the valve **30**. While the number of components is reduced, sealing of the ventilation passage **5** can be easily held by the valve **30**. Therefore, performance can be improved.

(40) In particular, in the present embodiment, the link **35** is directly joined to the fin **15** and the valve **30**. Therefore, a simpler configuration can be adopted, and assembly to the case body **3** is facilitated.

(41) The fin **15** is pivotable in the operation direction of the operation section **20** which causes the fin **15** to pivot. In this manner, the fin **15** can be intuitively operated by the operation of the operation section **20**, and when the fin **15** pivots in accordance with the operation of the operation section **20**, the valve **30** can pivot by being interlocked with the pivoting of the fin **15**. Therefore, the fin **15** and the valve **30** can be operated by the common operation section **20**, and thus, the configuration can be simplified.

(42) In addition, in a state where the ventilation passage 5 is closed by the valve 30, the air-conditioned wind does not flow through the ventilation passage 5. Therefore, fin 15 does not need to pivot. Therefore, even in a configuration in which the operation section 20 for causing the fin 15 to pivot is used to interlock the opening and closing of the valve 30, there is no inconvenience when the fin 15 is operated by the operation section 20.

(43) A configuration is adopted in which the link 35 is movable in a direction along the pivoting direction of the fin 15, and the valve 30 is caused to pivot by an external force applied to the link joining portion 38 from the link 35 side as the link 35 moves. In this manner, the pivoting of the valve 30 can be easily interlocked with the pivoting of the fin 15 by the link 35.

(44) In the link 35, the hole portion 37a of the valve side joining portion 37 joined to the link joining portion 38 is formed in a longitudinal shape in the direction intersecting with the longitudinal direction of the link joining portion 38. In this manner, so thin accordance with the length of the hole portion 37a, a timing at which an external force is transmitted from the fin 15 side to the link joining portion 38 of the valve 30 can be set by the link 35. Therefore, an operation amount of the operation section 20 required for the valve 30 to open and close the ventilation passage 5, that is, a swing angle of the fin 15 can be easily adjusted with a simple configuration.

(45) The plurality of fins 15 are disposed in the rightward-leftward direction which is the longitudinal direction of the case body 3, and the valve 30 is disposed in the longitudinal shape along the longitudinal direction. In this manner, the thin wind direction adjustment apparatus 1 having a thin shape can be configured.

(46) In this way, according to the present embodiment, it is possible to provide the wind direction adjustment apparatus 1 having a thin and miniaturized shape, and having a simple design.

(47) Next, embodiment 2 will be described with reference to FIGS. 5 and 9. The same reference numerals will be assigned to the same configurations and operations as those in the embodiment 1, and description thereof will be omitted.

(48) In the present embodiment, the link 35 is indirectly joined to the fin 15.

(49) In the shown example, in the link 35, the hole portion 37a of the valve side joining portion 37 is formed in a round hole shape which is slightly larger than an outer diameter dimension of the link joining portion 38. In addition, a joint 45 serving as a joining body is joined to the fin side joining portion 41. The link 35 is joined to the fin 15 via the joint 45.

(50) The joint 45 is located in the ventilation passage 5. In the joint 45, a rear end portion serving as one end portion has a first joint joining portion 47 which is a first joining body joining portion joined to the fin side joining portion 41 of the link 35 to be pivotable. One of the fin side joining portion 41 and the first joint joining portion 47 is a recessed portion, and the other is a projecting portion. In the present embodiment, the fin side joining portion 41 is formed as the recessed portion, and the first joint joining portion 47 is formed as the projecting portion having a spherical shape. The fin side joining portion 41 has a hole portion 41a penetrating in the upward-downward direction intersecting with or orthogonal to a penetrating direction of the hole portion 37a of the valve side joining portion 37.

(51) In addition, in the joint 45, a front end portion serving as the other end portion has a second joint joining portion 48 which is a second joining body joining portion joined to the link side joining receiving portion 42 of the fin 15 to be pivotable. One of the link side joining receiving portion 42 and the second joint joining portion 48 is a recessed portion or a hole portion, and the other is a projecting portion or a shaft portion. In the present embodiment, the link side joining receiving portion 42 is formed as the shaft portion having a columnar shape, and the second joint joining portion 48 is formed as the hole portion having a round hole shape. In the shown example, the link side joining receiving portion 42 is located on the pivoting axis of the fin 15. In the present embodiment, the link side joining receiving portion 42 is formed coaxially with the pivoting portion 16. The link side joining receiving portion 42 is located in a base end portion of the pivoting portion 16, and the pivoting portion 16 coaxially protrudes from the link side joining

receiving portion **42**.

(52) Furthermore, in the joint **45**, a cam joining portion **49** is formed between the first joint joining portion **47** and the second joint joining portion **48**. With reference to the cam joining portion **49**, the first joint joining portion **47** and the second joint joining portion **48** are located on sides opposite to each other. In a plan view, with respect to a virtual line connecting one of the first joint joining portion **47** and the second joint joining portion **48** to the cam joining portion **49**, the joint **45** is disposed so that the other of the first joint joining portion **47** and the second joint joining portion **48** is not located on the virtual line. That is, the joint **45** is formed in a curved shape.

(53) A first cam member **51** is attached to the cam joining portion **49**. The first cam member **51** has an opening portion **53**. The joint **45** is inserted into the opening portion **53**, and the first joint joining portion **47** is located on the upstream side with respect to the first cam member **51**, that is, on the rear side which is the valve **30** side. The second joint joining portion **48** is located on the downstream side with respect to the first cam member **51**, that is, on the front side which is the fin **15** side.

(54) The first cam member **51** has a cam joining receiving portion **54** that receives the cam joining portion **49** to be pivotable. One of the cam joining portion **49** and the cam joining receiving portion **54** is a recessed portion or a hole portion, and the other is a projecting portion or a shaft portion. In the present embodiment, the cam joining portion **49** is the shaft portion having a columnar shape, and the cam joining receiving portion **54** is the hole portion. The cam joining receiving portion **54** is located in the opening portion **53**.

(55) In addition, the first cam member **51** has a first cam portion **55** formed in the rear end portion which is one end portion, and a second cam portion **56** formed in the front end portion which is the other end portion. The first cam portion **55** and the second cam portion **56** are located to be separated from each other. The first cam portion **55** and the second cam portion **56** are located on sides opposite to each other with reference to the cam joining receiving portion **54**. In the present embodiment, the first cam portion **55** and the second cam portion **56** are shaft portions formed in a columnar shape. The first cam portion **55** and the second cam portion **56** are disposed parallel to each other. The first cam portion **55** and the second cam portion **56** are respectively located in the upper portion of the first cam member **51**.

(56) In the present embodiment, the first cam portion **55** is located on the left side of the first cam member **51**, and the second cam portion **56** is located on the right side of the first cam member **51**.

(57) The first cam portion **55** and the second cam portion **56** are connected to the second cam member **61**. The second cam member **61** is located above the first cam member **51** in the ventilation passage **5**. The second cam member **61** has a first cam receiving portion **63** and a second cam receiving portion **64** which receive the first cam portion **55** and the second cam portion **56**. According to a connection relationship between the first cam portion **55** and the second cam portion **56**, and the first cam receiving portion **63** and the second cam receiving portion **64**, a position where the joint **45** pivots by being interlocked with the fin **15**, that is, a pivoting range (pivoting angle) of the fin **15** in which an external force for causing the valve **30** to pivot is generated by the pivoting of the fin **15**, in other words, an operation range of the operation section **20** is set. The first cam member **51** and the second cam member **61** form a timing setting portion that sets a timing for interlocking the pivoting of the fin **15** and the pivoting of the valve **30** with each other.

(58) In the present embodiment, the first cam receiving portion **63** and the second cam receiving portion **64** are cam holes formed in the second cam member **61**, and into which the first cam portion **55** and the second cam portion **56** are inserted to restrict positions of the first cam portion **55** and the second cam portion **56**. The first cam receiving portion **63** and the second cam receiving portion **64** may pass penetrate the second cam member **61**, or may be formed as a recessed portion. In the present embodiment, the first cam receiving portion **63** is located on the left side of the second cam member **61**, and the second cam receiving portion **64** is located on the right side of the

second cam member **61**.

(59) The first cam receiving portion **63** is formed in a long hole shape in the forward-rearward direction. In an inner edge portion of the first cam receiving portion **63**, the rear end portion which is one end portion includes a first cam stopper portion **63a**. On both sides of the first cam stopper portion **63a**, and a first guide portion **63b** and a second guide portion **63c** continuously extend in the forward-rearward direction. A second cam stopper portion **63d** is formed in the front end portion of the first guide portion **63b** which is the other end portion. The second cam stopper portion **63d** and the second guide portion **63c** are connected by a continuous portion **63e**.

(60) The first cam stopper portion **63a** and the second cam stopper portion **63d** are stopper portions which receive the first cam portion **55** of the first cam member **51**.

(61) The first guide portion **63b** and the second guide portion **63c** are portions which guide the first cam portion **55** of the first cam member **51** as the first cam member **51** pivots. The first guide portion **63b** and the second guide portion **63c** are formed to be curved in an arc shape projecting leftward. In the shown example, the first guide portion **63b** continuously extends in the forward-rearward direction to the left side of the first cam stopper portion **63a**, and the second guide portion **63c** continuously extends in the forward-rearward direction to the right side of the first cam stopper portion **63a**. The first guide portion **63b** extends forward from the second guide portion **63c**.

(62) The second cam receiving portion **64** is formed in a long hole shape in the forward-rearward direction. The second cam receiving portion **64** is formed to be shorter in the forward-rearward direction than the first cam receiving portion **63**. In the inner edge portion of the second cam receiving portion **64**, the rear end portion which is one end portion includes a first cam stopper portion **64a**. A first guide portion **64b** continuously extends in the forward-rearward direction to the right side which is one side of the first cam stopper portion **64a**. A second guide portion **64c** continuously extends in the forward-rearward direction to the front end portion of the first guide portion **64b**. A second cam stopper portion **64d** is formed to continuously extend leftward in the front end portion of the second guide portion **64c** which is the other end portion. The second cam stopper portion **64d** and the first cam stopper portion **64a** are connected by a continuous portion **64e**. The first guide portion **64b** is formed to be curved in an arc shape projecting rightward. The first guide portion **64b** is located concentrically with the second guide portion **63c**. The second guide portion **64c** is formed in an arc shape projecting rearward.

(63) The first cam stopper portion **64a** and the second cam stopper portion **64d** are stopper portions which receive the second cam portion **56** of the first cam member **51**.

(64) The first guide portion **64b** and the second guide portion **64c** are portions which guide the second cam portion **56** of the first cam member **51** as the first cam member **51** pivots.

(65) The wind direction adjustment apparatus **1** of the present embodiment can blow the air-conditioned wind in any desired direction by combining the wind distribution of the fin **15** in the rightward-leftward direction and the wind distribution of the downstream side fin **24** in the upward-downward direction in accordance with the operation of the operation section **20**. The wind distribution in the upward-downward direction is the same as that in the embodiment 1, and thus, the description will be omitted.

(66) As shown in FIG. **8B**, when the operation section **20** is located in the central portion of the downstream side fin **24** in the rightward-leftward direction, that is, when the fin **15** is located at the neutral position, the air-conditioned wind straightly flows to the air outlet **9** along the flow straightening surface of the fin **15** inside the ventilation passage **5**. The air-conditioned wind blows in a front surface direction from the air outlet **9** along the axial direction of the case body **3**, that is, along the ventilation direction of the ventilation passage **5**.

(67) In addition, as shown in FIG. **8A**, when the operation section **20** is slid leftward along the downstream side fin **24** from the neutral position, the downstream side of the fin **15** joined to the operation section **20** moves leftward, and the fin **15** pivots in the clockwise direction around the pivoting portion **16** in the drawing. FIG. **8A** shows a state where the operation section **20** is slid

leftward to the maximum by being moved until the operation section **20** comes into contact with the restriction portion **25** on the left side, that is, a state where the fin **15** is swung leftward to the maximum (maximum leftward swing angle position, for example, 35° to the left). Therefore, the air-conditioned wind is straightened in the leftward direction along the flow straightening surface of the fin **15** inside the ventilation passage **5**, and blows in the leftward direction from the air outlet **9**.

(68) As the fin **15** pivots in the clockwise direction, the link side joining receiving portion **42** also moves in the clockwise direction around the pivoting portion **16**. Accordingly, the joint **45** in which the second joint joining portion **48** is joined to the link side joining receiving portion **42** tends to pivot in the clockwise direction, and the first cam member **51** tends to pivot in the counterclockwise direction. However, the pivoting of the first cam member **51** is restricted since the second cam portion **56** of the first cam member **51** is in contact with the first cam stopper portion **64a** of the second cam receiving portion **64** of the second cam member **61**. Therefore, the joint **45** held by the cam joining portion **49** serving as a pivoting center with respect to the first cam member **51** relatively pivots in the counterclockwise direction with respect to the downstream side of the fin **15** around the cam joining portion **49**. Therefore, the joint **45** maintains the same attitude and the same position as those when the fin **15** is located at the neutral position. In this manner, over the entire movement range, the link **35** does not apply an external force in the forward-rearward direction to the link joining portion **38** of the valve **30** inserted into the hole portion **37a** of the valve side joining portion **37**. Therefore, the valve **30** is not interlocked with the pivoting of the fin **15**, and the valve **30** maintains the open state of the ventilation passage **5**.

(69) On the other hand, as shown in FIG. **8C**, when the operation section **20** is slid rightward along the downstream side fin **24** from the neutral position, the downstream side of the fin **15** joined to the operation section **20** moves rightward, and the fin **15** pivots in the counterclockwise direction around the pivoting portion **16** in the drawing. Therefore, the air-conditioned wind is straightened rightward along the flow straightening surface of the fin **15** inside the ventilation passage **5**, and blows in the rightward direction from the air outlet **9**.

(70) As the fin **15** pivots in the counterclockwise direction, the link side joining receiving portion **42** also moves in the counterclockwise direction around the pivoting portion **16**. Accordingly, the joint **45** in which the second joint joining portion **48** is joined to the link side joining receiving portion **42** tends to pivot in the counterclockwise direction, and the first cam member **51** tends to pivot in the clockwise direction. With regard to the first cam member **51**, the first cam portion **55** moves to the first cam stopper portion **63a** along the second guide portion **63c** of the first cam receiving portion **63** of the second cam member **61**. When the second cam portion **56** pivots to move along the first guide portion **64b** of the second cam receiving portion **64** of the second cam member **61**, and the first cam portion **55** comes into contact with the first cam stopper portion **63a**, the pivoting of the first cam member **51** is restricted. Therefore, the joint **45** held by the cam joining portion **49** serving as the pivoting center with respect to the first cam member **51** relatively pivots in the clockwise direction with respect to the downstream side of the fin **15** around the cam joining portion **49**. Therefore, the joint **45** maintains the same attitude and the same position as those when the fin **15** is located at the neutral position. In this manner, over the entire movement range, the link **35** does not apply an external force in the forward-rearward direction to the link joining portion **38** of the valve **30** inserted into the hole portion **37a** of the valve side joining portion **37**. Therefore, the valve **30** is not interlocked with the pivoting of the fin **15**, and the valve **30** maintains the open state of the ventilation passage **5**.

(71) As shown in FIG. **8D**, when the operation section **20** is further slid rightward along the downstream side fin **24** from the state shown in FIG. **8C**, as the fin **15** further pivots, the first cam portion **55** of the first cam member **51** moves to the second cam stopper portion **63d** along the first guide portion **63b** of the first cam receiving portion **63** of the second cam member **61**, and the second cam portion **56** pivots to move to the second cam stopper portion **64d** along the second

guide portion 64c of the second cam receiving portion 64 of the second cam member 61. Therefore, the valve side joining portion 37 of the link 35 joined to the joint 45 interferes with the link joining portion 38 of the valve 30 in the forward-rearward direction, and pulls the link joining portion 38 to the downstream side, that is, to the front side. In this manner, the valve 30 pivots by being interlocked with the pivoting of the fin 15, and closes the ventilation passage 5 in accordance with the pivoting angle. FIG. 8D shows a state where the operation section 20 is slid rightward to the maximum by being moved until the operation section 20 comes into contact with the restriction portion 25 on the right side, that is, a state where the fin 15 is swung rightward to the maximum (maximum rightward swing angle position, for example, 82° to the right). At least in this state, the valve 30 is located at a position where the ventilation passage 5 is completely or substantially completely closed.

(72) In this way, according to the embodiment 2, the pivoting of the valve 30 disposed in the ventilation passage 5 in the case body 3 to be pivotable in the direction intersecting with the pivoting direction of the fin 15 is interlocked with the pivoting of the fin 15 disposed in the ventilation passage 5 in the case body 3 to be pivotable. The embodiment 2 has the same configuration as that of the embodiment 1. Accordingly, while the number of components can be reduced, performance can be improved. The embodiment 2 can achieve the same operational effects as those of the embodiment 1.

(73) In addition, the joint 45, the first cam member 51, and the second cam member 61 are used to set the timing at which the external force from the fin 15 side is transmitted to the link joining portion 38 of the valve 30 by the link 35. Therefore, the operation amount of the operation section 20, that is, the swing angle of the fin 15 required for the valve 30 to open and close the ventilation passage 5 can be finely adjusted in accordance with shapes of the first cam receiving portion 63 and the second cam receiving portion 64.

(74) In each of the above-described embodiments, a configuration is adopted in which the valve 30 is interlocked to pivot when the fin 15 is swung in the rightward direction by a predetermined angle or larger. However, without being limited thereto, the disposition or the shape of each portion may be inverted rightward and leftward. In this manner, the valve 30 may be interlocked to pivot when the fin 15 is swung in the leftward direction by a predetermined angle or larger.

(75) In addition, a configuration in which the wind direction adjustment apparatus 1 adopts the horizontal type has been described. However, the wind direction adjustment apparatus 1 can also be configured in the same manner even when a vertical type whose longitudinal direction is the upward-downward direction is adopted. In this case, a configuration is adopted in which the fin 15 is pivotable in the upward-downward direction, and the valve 30 is caused to pivot in the rightward-leftward direction. The disposition of the above-described respective embodiments is rotated by 90° when viewed from the front surface. In this manner, it is possible to achieve the same operational effects as those of the above-described respective embodiments.

(76) Furthermore, the wind direction adjustment apparatus 1 is not limited to an apparatus for an automobile, and may be used for any other desired purpose.

INDUSTRIAL APPLICABILITY

(77) For example, the present invention can be suitably used as a wind direction adjustment apparatus for air conditioning of an automobile.

REFERENCE SIGNS LIST

(78) 1 Wind direction adjustment apparatus 3 Case body 5 Ventilation passage 15 Fin 20 Operation section 30 Valve 35 Link 38 Link joining portion

Claims

1. A wind direction adjustment apparatus comprising: a case body that internally defines a ventilation passage; a fin disposed in the ventilation passage in the case body to be pivotable to

adjust a wind direction in accordance with pivoting; a valve disposed in the ventilation passage in the case body to be pivotable in a direction intersecting with a pivoting direction of the fin, and capable of opening and closing the ventilation passage in accordance with pivoting; a link that interlocks the pivoting of the valve with the pivoting of the fin; and an operation section that causes the fin to pivot, wherein the fin is pivotable in an operation direction of the operation section, and the link is movable in a direction along the operation direction of the operation section that causes the fin to pivot.

2. The wind direction adjustment apparatus according to claim 1, wherein the link is movable in a direction along the pivoting direction of the fin, and the valve has a link joining portion joined to the link, and is caused to pivot by an external force applied to the link joining portion from a link side as the link moves.

3. The wind direction adjustment apparatus according to claim 2, wherein a plurality of the fins are disposed in a longitudinal direction of the case body, and the valve is disposed in a longitudinal shape along the longitudinal direction.

4. The wind direction adjustment apparatus according to claim 1, wherein a plurality of the fins are disposed in a longitudinal direction of the case body, and the valve is disposed in a longitudinal shape along the longitudinal direction.
